TEST CASE 1: RETAIL SALES PERFORMANCE ANALYSIS

```
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Create dataset
data = {
   'Product': ['Laptop', 'Mobile', 'Tablet', 'Laptop', 'Mobile', 'Tablet'],
   'Quantity': [5, 10, 7, 3, 5, 2],
   'Price': [50000, 15000, 20000, None, 16000, 21000]
}

# Step 2: Create DataFrame
df = pd.DataFrame(data)

# Step 3: Handle missing values
df['Price'] = df.groupby('Product')['Price'].transform(
   lambda x: x.fillna(x.mean())
)

# Step 4: Calculate total sales
df['Total_Sales'] = df['Quantity'] * df['Price']

# Step 5: Find product with highest sales
top_product = df.groupby('Product')['Total_Sales'].sum().idxmax()

print("Product with highest total sales:", top_product)

# Step 6: Display sales summary
sales_summary = df.groupby('Product')['Total_Sales'].sum()
```

```
print("\nTotal Sales by Product:")
print(sales_summary)

# Step 7: Plot bar chart
sales_summary.plot(
   kind='bar',
   color='skyblue',
   figsize=(8, 5)
)

plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

TEST CASE 2: CITY TEMPERATURE MONITORING

```
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Create dataset
data = {
   'Date': ['2025-05-10', '2025-05-11', '2025-05-12',
         '2025-05-13', '2025-05-14', '2025-05-15'],
   'Min_Temp': [25, 26, None, 24, 27, 28],
   'Max_Temp': [35, 36, 34, None, 38, 40]
}

# Step 2: Create DataFrame
df = pd.DataFrame(data)

# Step 3: Handle missing values using mean
df['Min_Temp'] = df['Min_Temp'].fillna(df['Min_Temp'].mean())
df['Max_Temp'] = df['Max_Temp'].fillna(df['Max_Temp'].mean())

# Step 4: Calculate average temperature
df['Average_Temp'] = (df['Min_Temp'] + df['Max_Temp']) / 2

# Step 5: Find the hottest date
hottest = df.loc[df['Average_Temp'].idxmax(), 'Date']

print("Date with highest average temperature:", hottest)

# Step 6: Display temperature data
print("\nTemperature Summary:")
print(df.to_string(index=False))
```

```
# Step 7: Plot average temperature
plt.figure(figsize=(9, 5))

plt.plot(
   df['Date'],
   df['Average_Temp'],
   marker='o',
   color='orange',
   linewidth=2
)

plt.xlabel('Date')
plt.ylabel('Average Temperature (°C)')
plt.title('Average Temperature Over Time')

plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()

# Step 8: Display graph
plt.show()
```

TEST CASE 3: DAILY TRAFFIC VOLUME REPORT

```
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Create dataset
data = {
   'Date': [
      '2025-04-10', '2025-04-11', '2025-04-12',
      '2025-04-13', '2025-04-14', '2025-04-15'
   ],
   'Vehicle_Count': [500, 450, 800, None, 600, 550]
}

# Step 2: Create DataFrame
df = pd.DataFrame(data)

# Step 3: Handle missing values
df['Vehicle_Count'] = df['Vehicle_Count'].fillna(0)

# Step 4: Calculate total and average vehicles
total = df['Vehicle_Count'].sum()
avg = df['Vehicle_Count'].mean()

print(f"Total vehicles: {total:.0f}")
print(f"Average daily vehicles: {avg:.2f}")

# Step 5: Find date with highest traffic
peak = df.loc[df['Vehicle_Count'].idxmax(), 'Date']

print(f"Date with highest traffic: {peak}")
```

```
# Step 6: Display traffic data
print("\nTraffic Summary:")
print(df.to_string(index=False))

# Step 7: Plot daily traffic volume
plt.figure(figsize=(9, 5))

plt.plot(
   df['Date'],
   df['Vehicle_Count'],
   marker='o',
   color='red',
   linewidth=2
)

plt.xlabel('Date')
plt.ylabel('Vehicle Count')
plt.title('Daily Traffic Volume Trend')

plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()

# Step 8: Display graph
plt.show()
```

TEST CASE 4: E-COMMERCE PRODUCT RATINGS

```
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Create dataset
data = {
   'Product_Name': [
      'Laptop', 'Headphones', 'Smartwatch',
      'Camera', 'Phone', 'Monitor'
   ],
   'Customer_Rating': [4.8, 4.5, None, 4.7, 4.9, 4.6]
}

# Step 2: Create DataFrame
df = pd.DataFrame(data)

# Step 3: Remove missing ratings
df = df.dropna(subset=['Customer_Rating'])

# Step 4: Calculate average rating
avg = df['Customer_Rating'].mean()

print(f"Average Rating: {avg:.2f}")

# Step 5: Find top 3 products
top = df.sort_values(
   'Customer_Rating',
   ascending=False
).head(3)
```

```
print("\nTop 3 Products:")
print(top[['Product_Name', 'Customer_Rating']].to_string(index=False))

# Step 6: Display top 5 products using bar chart
top5 = df.sort_values(
   'Customer_Rating',
   ascending=False
).head(5)

plt.figure(figsize=(9, 5))

top5.plot(
   kind='bar',
   x='Product_Name',
   y='Customer_Rating',
   color='teal',
   legend=False,
   figsize=(9, 5)
)

plt.xlabel('Product Name')
plt.ylabel('Customer Rating')
plt.title('Top 5 Product Ratings')
plt.ylim(0, 5)
plt.xticks(rotation=45)
plt.tight_layout()

# Step 7: Show graph
plt.show()
```

TEST CASE 5: STUDENT COURSE COMPLETION REPORT

```
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Create dataset
data = {
   'Student_ID': ['S001', 'S002', 'S003', 'S004', 'S005', 'S006'],
   'Completion_Status': ['Yes', 'No', None, 'Yes', 'No', 'Yes']
}

# Step 2: Create DataFrame
df = pd.DataFrame(data)

# Step 3: Handle missing values
df['Completion_Status'] = df['Completion_Status'].fillna('No')

# Step 4: Count completion status
count = df['Completion_Status'].value_counts()

print("Course Completion Status:")
print(count)

# Step 5: Display pie chart
plt.figure(figsize=(6, 6))

count.plot(
   kind='pie',
   autopct='%1.1f%%',
   colors=['lightgreen', 'lightcoral'],
   startangle=90
```

```
)

plt.title('Course Completion Status')
plt.ylabel('')

# Step 6: Show chart
plt.tight_layout()
plt.show()
```